Lecture notes: correlations near quantum criticality

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Contents

1 Quick reminders about finite-temperature formalism **2**

2 Concrete issues about quantum phase transitions **3**
2.1 Quantum phases of the Ising chain . . . 3
2.2 Quantum criticality . . . 3
2.3 Field theory approach and the question of anomalous dimension . . . 3

3 Fermionization and exact results **4**
3.1 Jordan-Wigner transformation . . . 4
3.2 Reformulating the question of anomalous dimension . . . 4
3.3 Exact results, and the point of this lecture . . . 5

4 Approach from the ordered phase **6**
4.1 Classical picture and justification . . . 6
4.2 Computing the finite-T static correlator . . . 6
4.3 From finite-T static to dynamical correlator . . . 6

5 Approach from the quantum disordered phase **7**
5.1 The nature of elementary excitations . . . 7
5.2 The static $T = 0$ correlator: exact result and phenomenological picture . . . 8
5.3 From static to dynamical at $T = 0$ . . . 9
5.4 From $T = 0$ to finite temperature . . . 9

6 The quantum critical region **10**
6.1 Basic correlator properties . . . 10
6.2 From zero to finite T using the conformal map . . . 10

Problem sheet – quantum phase transition of free fermions **12**

Solution sheet – quantum phase transition of free fermions **13**

Prerequisites

Before embarking on this course, it is recommended that you have some basic knowledge about:

- many-body theory at finite temperature,
- statistical field theory and renormalization,
- quantum mechanics.

Introduction

These notes are a transcript of a lecture I gave last summer at the Technical University of Munich, in the context of the "Advanced Methods in Quantum Many-Body Theory" course. My lecture was part of a series of lectures on quantum criticality that covered various aspects of the problem, this is why I am only treating a rather specific angle, which is that of computing correlation functions across the finite-temperature phase diagram of the quantum Ising model, using phenomenological and semiclassical arguments. I later wrote these notes as a standalone, removing explicit references to the other parts of the course as it actually took place, but not adding any further material that would make the lecture less specialized. These notes are largely inspired by Subir Sachdev's book on quantum phase transitions, especially Chapter X thereof.

1 Quick reminders about finite-temperature formalism

This lecture is devoted to studying correlation functions of quantum systems at finite temperature. In the following we will call them $C(x,t) = \langle \hat{\sigma}^z(x,t)\,\hat{\sigma}^z(0,0)\rangle$ at finite temperature. We will also encounter $S(k,\omega)$ the dynamical structure factor, and $\chi(k,\omega)$ the dynamical susceptibility, both of which are directly related to $C(x,t)$. The dynamical structure factor is the direct Fourier transform of the correlator,

$$S(k,\omega) = \int \mathrm{d}t \int \mathrm{d}x\, e^{i(\omega t - kx)} C(x,t). \tag{1}$$

Meanwhile, the dynamical susceptibility $\chi(k,\omega)$, that is a thermodynamical property, is the analytical continuation to the real frequency axis, $i\omega_n \to \omega$, of

$$\chi(k,i\omega_n) = \int_0^{1/T} \mathrm{d}\tau \int \mathrm{d}x\, e^{i(\omega_n \tau - kx)} C(x,-i\tau), \tag{2}$$

where $C(x,-i\tau)$ is the analytic continuation of $C(x,t)$ to the imaginary time axis. The whole situation is summarized in Fig.1.

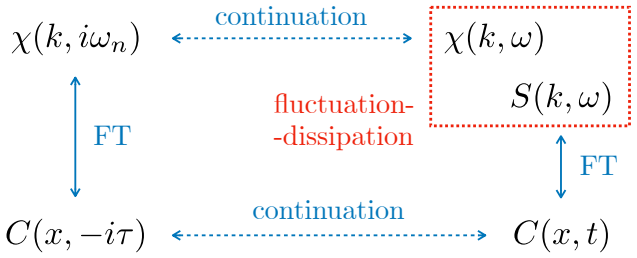


Figure 1: Summary of relations between various quantities related to a system's finite-temperature response.

There are quite a few useful things to remember about this, and the most urgent lesson is a warning: *Fourier transform and analytic continuation do not naively commute*. Indeed, $S(k,\omega) \neq \chi(k,\omega)$. Instead they are related by fluctuation-dissipation:

$$S(k,\omega) = 2\,(1 + n_{\mathrm{B}}(\omega/T))\ \mathrm{Im}\chi(k,\omega). \tag{3}$$

Beyond this trap that just requires you to remember about it, there is a real concrete issue: asking about $T > 0$ dynamical correlators is a highly nontrivial question. *In principle*, as we discussed above, it suffices to consider the euclidean action in $D = d+1$ space-time dimensions (with $\tau \in [0, 1/T]$) and compute imaginary-time correlators, then just analytically continue those to real times. *In practice*, however, this analytic continuation is generally very badly behaved numerically, and comes in useful basically only if one has an exact solution. Fortunately, as far as this lecture is concerned we will be in the latter case.

2 Concrete issues about quantum phase transitions

2.1 Quantum phases of the Ising chain

Considering a quantum system at finite temperature, for instance the paradigmatic quantum Ising chain ($d = 1$) in a transverse field

$$H_{\text{Ising}} = -J \sum_i \left(\hat{\sigma}_i^z \hat{\sigma}_{i+1}^z + g \hat{\sigma}_i^x \right), \tag{4}$$

let us first ask what are the phases of this system at $T = 0$. For concreteness we take $J > 0$ and $g > 0$. In the limit $g \to \infty$, the system is a quantum paramagnet, as the ground state is that where all spins point along the $+x$ direction (parallel to the transverse magnetic field). Elementary excitations are spin flips, where one single spin locally points in the $-x$ direction, with energy $2Jg$. In the limit $g \to 0$, the system spontaneously orders, as the ground state is that where all spins point in the $\pm z$ direction, with a twofold degeneracy. Elementary excitations are domain walls, that separate two ground state configurations, with energy $2J$.

2.2 Quantum criticality

Between the two limits $g \to 0$ and $g \to \infty$, there is a quantum critical point at some intermediate value $g_\star$, that separates the two phases. Quite generally in the whole parameter space (here, of values of g), there must be a function $\Delta(g)$ whose absolute value $|\Delta|$ sets the typical energy scale separating the ground state from excited states, that vanishes at the quantum critical point, $\Delta(g_\star) = 0$, reflecting the fact that critical fluctuations occur at all scales.

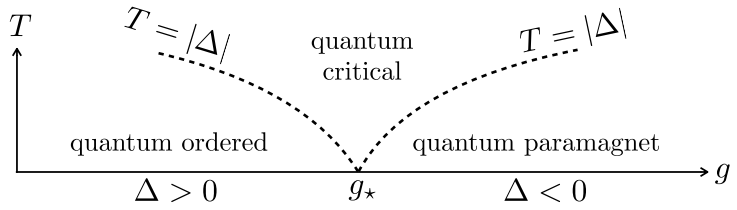


Figure 2: Schematics of the situation relevant for the quantum Ising chain at finite temperature.

In this lecture, we will be interested in the dynamical correlation functions at finite temperature $T > 0$, and we will be asking about correlation functions' scaling as a function of T/Δ. We will focus on a case where the system has a phase transition only at $T = 0$, while at $T > 0$ it has only a disordered phase (as per Peierls' argument since we are in $d = 1$). Therefore, the general idea of the phase diagram is that of Fig.2. While at low temperatures the system's behavior may not be significantly different from the $T = 0$ characterization, at large enough temperatures, namely $T \gtrsim |\Delta|$, thermal fluctuations become important and the physics is that of a "quantum critical phase" that we will see is governed by the QCP.

2.3 Field theory approach and the question of anomalous dimension

We want to understand the critical properties of a system that has a phase transition between an ordered and a disordered phase. A natural approach would be to use a mapping to a bosonic field theory, where the boson ϕ identifies as the field that undergoes ordering, $\hat{\sigma}^z \leftrightarrow \phi$. This can be made more precise, but here we can stay phenomenological and guess that the resulting theory will be gaussian (à la Ginzburg-Landau),

$$\mathcal{Z} = \int \mathrm{D}\phi \, e^{-S[\phi]}, \qquad S[\phi] = \int \mathrm{d}^d x \mathrm{d}\tau \, \frac{1}{2} \left((\partial_\tau \phi)^2 + c^2 (\partial_x \phi) + r \phi^2 \right) + \frac{u}{4!} (\phi^2)^2, \tag{5}$$

identifying $r = g - g_\star$ to connect with the previous subsection.

For now take $u = 0$ and discuss the behavior of operators under renormalization (i.e, their bare behavior). By definition, following the Kadanoff coarse-graining picture, $\dim[x] = -1$, and the dynamical exponent z is defined through $\dim[\tau] = -z$. That is to say, the renormalization flow of coordinates is $x' = xe^{-\ell}$ and $\tau' = \tau e^{-z\ell}$

as $\ell \to +\infty$. Invariance of the non-interacting terms (keeping c non flowing) requires $z = 1$ and transformation of the fields as

$$\phi' = \phi e^{\ell(D-2)/2}, \tag{6}$$

i.e. $\dim[\phi] = (D-2)/2$, whence $\dim[r] = +2$. Obviously r is relevant, as it is the parameter that drives the quantum phase transition.

Now, our main object of interest is $\hat{\sigma}^z$ equivalently ϕ, and we want to know the dimension of that object in the actual model Eq.(4) equivalently Eq.(5) with finite interactions $u > 0$. Clearly, $\dim[u] = 4 - D$, so u is relevant in low enough dimensions – especially in our case, $D = 2$. The fixed point at $(r, u) = (0, 0)$ is unstable, and we know we should look for the Wilson-Fisher fixed point. There, fields are known to have an anomalous dimension,

$$\dim[\phi] = (D - 2 + \eta)/2 \tag{7}$$

where η is zero above the upper critical dimension ($D = 4$), but can be nonzero otherwise. This coefficient η is extremely important, because it determines the critical behavior of correlation functions, as we will show later. One can access it approximately, for instance using the $D = 4 - \epsilon$ expansion and diagrammatics in powers of u, but this is clearly not exact and for the Ising chain it looks a bit dubious since one needs to extend down to $D = 2$. So, ideally one would look for a better approach.

3 Fermionization and exact results

3.1 Jordan-Wigner transformation

For $d = 1$ quantum spin systems (with a 2-dimensional Hilbert space per site), it is possible to fermionize the model by means of the Jordan-Wigner transformation. No details will be provided here. The result of the procedure is the mapping

$$H_{\text{Ising}} \mapsto \int \frac{\mathrm{d}k}{2\pi} \, \psi_k^\dagger \left(\xi_k \tau^z + \Delta_k \tau^y \right) \psi_k, \tag{8}$$

$$\psi_k = (c_k, c_{-k}^\dagger), \qquad \xi_k = 2J \sin(ka), \Delta_k = 2J(1 - g\cos(ka)), \tag{9}$$

where a is the lattice spacing. Here we will use an approach via continuum (long-wavelength) field theory, so just consider $\xi_k \simeq ck$ with $c = 2Ja$ and $\Delta_k \simeq \Delta = 2J(1-g)$, so that the field theory in real space reads

$$H = \int \mathrm{d}x \left(\tfrac{1}{2} c(\psi^\dagger \partial_x \psi^\dagger - \psi \partial_x \psi) + \Delta \psi^\dagger \psi \right). \tag{10}$$

The behavior of Δ and the phase diagram are those of Fig.2, with a spectrum of excitations $\epsilon_k = \sqrt{c^2 k^2 + \Delta^2}$ and a QCP at $g_\star = 1$ where the gap closes. At criticality the dispersion is linear, which is related to $z = 1$ and to conformal symmetry, see later.

3.2 Reformulating the question of anomalous dimension

The action, in this fermionic language, now reads

$$S[\psi^\dagger, \psi] = \int \mathrm{d}\tau \left(\mathrm{d}x \, \psi^\dagger \partial_\tau \psi - H \right). \tag{11}$$

How do the model's parameters flow under renormalization? Again transform the coordinates as $x' = xe^{-\ell}$ and $\tau' = \tau e^{-z\ell}$: a fixed point at $\Delta = 0$ is guaranteed provided $z = 1$ (fortunately that hasn't changed) and transforming the fields as $\psi' = \psi e^{\ell/2}$ in other words $\dim[\psi] = 1/2$. Away from $\Delta = 0$ it is clear that $\dim[\Delta] = +1$: the gap is of course a relevant perturbation. Meanwhile $\dim[c] = 0$: this is a scale-invariant parameter of the theory.

So, what have we gained? Well, contrarily to the bosonic attempt Eq.(5), this critical fermionic theory is nice because it's stable, in the sense that all perturbations are irrelevant. Indeed, for fermions one cannot write

a local term like $(\psi^\dagger\psi)^2$, because of Pauli exclusion [1]. Instead, any interaction term needs to involve more derivatives, for instance $(\psi^\dagger\partial_x\psi)^2$ or worse, and this ensures that all possible couplings have negative dimension i.e. they are irrelevant. So Eq.(11) is a universal description of criticality in $H_{\rm Ising}$.

That said, have we solved everything? Not quite. For instance we have not yet solved the question of the anomalous dimension $\dim[\hat\sigma^z]$. This remains a nontrivial question, because the Jordan-Wigner mapping $\hat\sigma^z \leftrightarrow \psi$ is complicated (in particular, nonlocal). Thus, even though it's a very simple matter to derive the correlations between our free fermions,

$$\langle\psi(x,-i\tau)\psi^\dagger(0,0)\rangle = \frac{1}{4\pi}\left(\big[c\tau - ix\big]^{-1} + \big[c\tau + ix\big]^{-1}\right), \qquad T=0, \tag{12a}$$

$$\langle\psi(x,-i\tau)\psi^\dagger(0,0)\rangle = \frac{1}{4\pi}\frac{\pi T}{c}\left(\big[\sin((\pi T/c)(c\tau - ix))\big]^{-1} + \big[\sin((\pi T/c)(c\tau + ix))\big]^{-1}\right), \qquad T>0, \tag{12b}$$

it's still hard to get the correlations between spins $\hat\sigma^z$. In particular, there is no easy way to deduce $\dim[\hat\sigma^z]$ directly from $\dim[\psi]=1/2$.

3.3 Exact results, and the point of this lecture

While computing the dynamical correlator $C(x,-i\tau) = \langle\hat\sigma^z(x,-i\tau)\hat\sigma^z(0,0)\rangle$ exactly remains very hard or impossible, there are ways to compute the *static* correlators of fermionic chains such as Eq.(11). This machinery, which I will not discuss here, involves Pfaffians, Toeplitz determinants, etc, and provides us with an exact asymptotic result:

$$C(x,0) \overset{|x|\to\infty}{=} Z\,T^{1/4}\,G(\Delta/T)\,\exp\left[-(T/c)\,|x|\,F(\Delta/T)\right]. \tag{13}$$

Here Z is a non-universal constant that depends on details of the model (actually $Z = J^{-1/4}$), and F and G are two real functions that are known exactly, but for our purposes it is sufficient to look at their asymptotics that are shown in Fig.3.

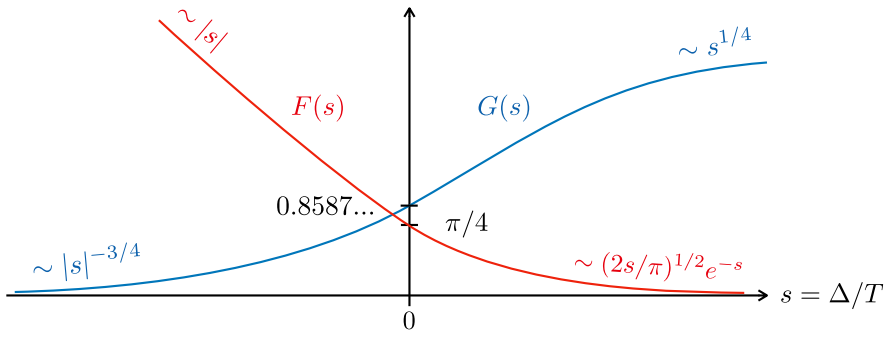


Figure 3: Asymptotics of the $F(s)$ and $G(s)$ functions that govern static correlators, Eq.(13).

Note that because $\dim[T] = z = +1$ (obviously temperature is a relevant quantity), the $T^{1/4}$ prefactor indicates that $\dim[\hat\sigma^z] = 1/8$. Equivalently, $\eta = 1/4$. This is the anomalous dimension, and it is an exact nonperturbative result ! It is also related to the critical exponent of magnetization, $m \propto (T_{\rm c} - T)^{1/8}$ in Onsager's solution of the classical $D=2$ Ising model.

This is of course very satisfactory, but we still do not have the dynamical correlators $C(x,t)$. The plan for this lecture will be to recover the static result Eq.(13) for $C(x,0)$, and to go further and obtain the dynamical function $C(x,t)$, by means of *classical* methods that will prove both powerful and physical.

[1] This is only true for our *flavorless* JW fermions, due to the original local Hilbert space dimension being 2.

4 Approach from the ordered phase

4.1 Classical picture and justification

Let us consider the phase $\Delta > 0$ in the limit $\Delta \gg T$. Then, asymptotics of Eq.(13) give us

$$C(x,0) \overset{|x|\to\infty}{=} n_0^2\, e^{-|x|/\xi}, \tag{14}$$

that at $T = 0$ describes long-range order with order parameter $n_0^2 = Z\,\Delta^{1/4}$, and displaying a correlation length

$$\xi = \left(\frac{\pi c^2}{2T\Delta}\right) e^{\Delta/T}, \tag{15}$$

that unsurprisingly diverges exponentially at $T \to 0$.

We will want to recover these results, Eqs.(24)–(15), by classical arguments. Why can one expect this to be even possible? As is well known for instance when discussing quantum vs classical gases, one needs to compare the particles' thermal de Broglie length $\lambda_{\rm dB}$ with the typical distance between them. Here, the elementary excitations in the ordered phase (domain walls) have kinetic energy $E - \Delta \approx c^2\lambda_{\rm dB}^2/2\Delta$, where $\lambda_{\rm dB}$ is set by equating $E - \Delta \equiv T$, whence the typical domain wall size $\lambda_{\rm dB} = c/\sqrt{\Delta T}$. Meanwhile the typical distance between domain walls is simply the correlation length ξ. So we look at the ratio $\xi/\lambda_{\rm dB} = \sqrt{\pi/2}\, e^{\Delta/T} \gg 1$: this phase is well described as a classical gas of domain walls.

4.2 Computing the finite-T static correlator

From the energy of domain walls $\epsilon_k \approx \Delta + c^2k^2/2\Delta$, one can evaluate their classical Maxwell-Boltzmann density,

$$\rho = \int \frac{\mathrm{d}k}{2\pi} e^{-\epsilon_k/T} = \left(\frac{T\Delta}{2\pi c^2}\right)^{1/2} e^{-\Delta/T}. \tag{16}$$

Now let us consider a system of size L and aim at computing $C(x,0)$. The total number of particles is $N = \rho L$ and there is a probability $q = |x|/L$ that a *given* particle will be found within $[0,x]$ – this is because classical particles can be counted with independent probabilities. Then, a given *distribution* of n tagged particles within $[0,x]$ and $N-n$ tagged particles outside of $[0,x]$ has probability $q^n(1-q)^{N-n}$, and removing the tags there are $\binom{N}{n}$ such distributions. Each of these distributions contributes an amount $n_0^2(-1)^n$ to the static correlation $C(x,0)$, because each one of the n domain walls flips the spin once. This situation is depicted in Fig.4.

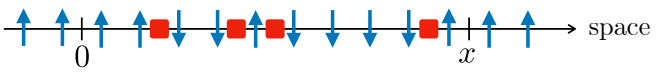


Figure 4: Blue arrows: spins (up or down). Red squares: particles (= domain walls). Here there are $n = 4$ particles within $[0,x]$.

Summing over all contributions weighted by their probabilities, one finds

$$C(x,0) = n_0^2 \sum_{n=0}^{N} (-1)^n q^n (1-q)^{N-n} \binom{N}{n} = n_0^2 (1-2q)^N \simeq n_0^2\, e^{-2\rho|x|}. \tag{17}$$

Thus, we identify $\xi^{-1} = 2\rho$, which is precisely the exact asymptotic result we endeavoured to recover !

4.3 From finite-T static to dynamical correlator

For the calculation of dynamical correlators, we will use the same reasoning as above, but now in a $D = 2$ dimensional picture with a time axis. Now, each domain wall has a life-line, with velocity $v_k = c^2k/\Delta$: this is depicted in Fig.5. The reasoning is the same as previously, only now one has to draw a line[2] from $(0,0)$ to (x,t)

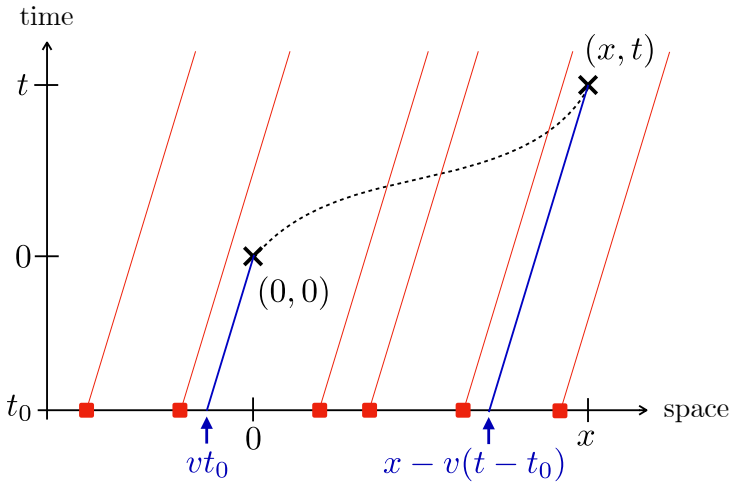


Figure 5: Red lines: life-lines at a given fixed velocity v (their slope) of all particles (red squares) present in the system. The number of such particles crossing the dashed black line can be traced back to the number of particles within the region of size $x - vt$ between the two blue arrows on the horizontal axis (here $n = 3$).

and count the number of life-lines (still representing domain walls, so each one inducing a spin flip) crossing it.

The density of particles with a given velocity v_k crossing that line is not $q = |x|/L$ but actually $|x - v_k t|/L$, see the figure. So we simply need to adapt the calculation Eq.(17) with this setting at finite v_k, and average over momenta. We obtain

$$C(x,t) = n_0^2\, R(x,t), \tag{18a}$$

$$R(x,t) = \exp\left\{-2\int \frac{\mathrm{d}k}{2\pi}\, e^{-\epsilon_k/T}\, |x - v_k t|\right\}. \tag{18b}$$

So we have achieved a great victory. Not only did we recover the exactly known static correlator $C(x,0)$, at least at long enough distances, using purely classical arguments. We also just derived with our classical arguments the *dynamical* correlator $R(x,t)$, that was not accessible to the exact derivation mentioned above; and comparison with numerical simulations of classical Ising spins shows basically perfect agreement.

In particular let us look at the local correlator $R(0,t) = e^{-|t|/\tau_\varphi}$, the calculation gives

$$\tau_\varphi^{-1} = \frac{2}{\pi}\int \mathrm{d}k \frac{\mathrm{d}\epsilon_k}{\mathrm{d}k} e^{-\epsilon_k/T} = \frac{2}{\pi}\int_\Delta^\infty \mathrm{d}\epsilon\, e^{-\epsilon/T} = \frac{2T}{\pi}\, e^{-\Delta/T}, \tag{19}$$

τ_φ is the quantum coherence time. Just like the correlation length ξ, it diverges in the limit $T \to 0$.

More generally it is easy (yet still non trivial) to understand that at $\Delta \ll T$, the full $R(x,t)$ only depends on the rescaled coordinates x/ξ and t/τ_φ.

5 Approach from the quantum disordered phase

5.1 The nature of elementary excitations

We now consider the case $\Delta < 0$ in the limit $|\Delta| \gg T$. Then the exact asymptotic result Eq.(13) tells us the inverse correlation length is $\xi^{-1} = |\Delta|/c$, finite: this is consistent with short-range correlations in a paramagnetic phase. We have already seen that deep in that phase, elementary excitations are local spin-flips, with energy $2gJ$, with one-particle wavefunctions being plane waves,

$$|k\rangle = \sum_j e^{ikx_j}|j\rangle, \qquad |j\rangle \equiv \big|\uparrow\cdots\uparrow_{j-1}\downarrow_j\uparrow_{j+1}\cdots\uparrow\big\rangle, \tag{20}$$

[2]It does not matter that the line be straight, describe a bijection $x \leftrightarrow t$, or even be non-self-intersecting, since the number of crossings modulo 2 is an invariant under any deformations of that line.

where up and down arrows refer to the $\pm x$ orientation. When these are dilute, the physics is that of a classical gas of particles. What is the precise nature of these particles? It is easy to see these are hard-core bosons with local (contact) repulsion. Indeed, if we project the Hamiltonian Eq.(4) onto the space of two-particle states (that are eigenstates of the $\hat{\sigma}^x$ bit), it reads

$$\mathcal{P}_{2\mathrm{P}} H_{\mathrm{Ising}} \mathcal{P}_{2\mathrm{P}} = -J \sum_i \left(\hat{\sigma}_i^+ \hat{\sigma}_{i+1}^- + \hat{\sigma}_i^- \hat{\sigma}_{i+1}^+ \right) + 4Jg, \tag{21}$$

that constitutes a repulsion energy of $+2J$ when two spins are flipped on neighboring sites $(i, i+1)$.

A consequence of this, and an ingredient we will need later, is that the S-matrix (a two-particle property) is particularly simple and universal in $d = 1$, and makes hard-core bosons look very much like fermions. Let us see this by writing a two-particle wavefunction

$$|k_1, k_2\rangle = \sum_{i,j} \left(e^{i(k_1 x_i + k_2 x_j)} + S_{k_1,k_2}\, e^{i(k_2 x_i + k_1 x_j)} \right) |i, j\rangle, \tag{22}$$

valid for x_i, x_j far enough from each other. This form is very particular and specific to $d = 1$, where conservation of total energy and total momentum upon scattering implies that the initial and final momenta (k_1, k_2) are either conserved, or exchanged only with a phase factor S_{k_1,k_2}. Now what is this phase factor? The scattering event is a two-body probem that can be solved in the center of mass frame, writing $x = x_i - x_j$: just write Schrödinger's equation

$$\left(-\frac{\mathrm{d}^2}{\mathrm{d}x^2} + U\, \delta(x) \right) \psi(x) = E\, \psi(x) \tag{23}$$

where $U\,\delta(x)$ models a contact repulsion. Dimensionally, the S-matrix can only be a function of U/k where $k = k_1 - k_2$. We are asking about the long-wavelength limit $k \to 0$ which is thus equivalent to asking about $U \to \infty$. Then we have an infinite wall at $x = 0$ which imposes that the wavefunction should vanish there: this imposes the important result $S_{k_1,k_2} = -1\ \forall k_1, k_2$.

5.2 The static $T = 0$ correlator: exact result and phenomenological picture

Recall that our main goal is to compute $C(x, t)$ at $T > 0$. Similar to before, asymptotics of Eq.(13) now give us

$$C(x, 0) \overset{|x|\to\infty}{=} \frac{ZT}{|\Delta|^{3/4}}\, e^{-|x|/\xi}, \qquad T > 0, \tag{24}$$

but it is not clear a priori how one can access $C(x, t)$ from there. What strategy should we adopt? Here, in fact, it will be useful to first go to $T = 0$, and to obtain a phenomenological derivation for the $T = 0$ static correlator $K(x, 0)$. Then, armed with that physical intuition, we will derive the dynamical $T = 0$ correlator $K(x, t)$, then eventually go back to finite temperature.

The physical idea is that elementary excitations in the paramagnetic phase (spin flips) are well described as quasiparticle excitations of the gaussian theory $S[\phi]$, recall Eq.(5). Essentially this means we can identify $\hat{\sigma}^z(x, t) = \sqrt{A}\, \phi(x, t)$, where A is the quasiparticle weight (that we will fix later). The propagator of our particles, at $T = 0$, is the paramagnet's dynamical susceptibility

$$\chi(k, \omega) = \frac{A}{c^2 k^2 + \Delta^2 - (\omega + i0)^2} + \ldots, \tag{25}$$

where "$+\ldots$" contains contributions from the continuum of many-particle states. From this we have

$$K(x, 0) = \int \frac{\mathrm{d}\omega}{2\pi} \int \frac{\mathrm{d}k}{2\pi}\, e^{-ikx} \chi(k, i\omega) = \frac{1}{2} \int \frac{\mathrm{d}k}{2\pi}\, e^{-ikx} \frac{A}{\epsilon_k} = \frac{A}{2\pi c}\, \mathsf{K}_0(|\Delta||x|/c) \overset{|x|\to\infty}{=} \frac{A}{\sqrt{8\pi c |\Delta||x|}} e^{-|x||\Delta|/c}, \tag{26}$$

where $\epsilon_k = \sqrt{\Delta^2 + c^2 k^2}$, the i in $i\omega$ is important, and K_0 is the modified Bessel function.

To fix A, let's go to finite temperature. Notice that the $T \to 0$ and $|x| \to \infty$ limits do not commute, which is apparent because the prefactors of $|x|$ and T in Eqs.(24),(26) are different. So it is not possible to just crank down $T \to 0$ in Eq.(13). What can be done, however, is replacing the energy integral with a Matsubara sum,

$$\int \frac{\mathrm{d}\omega}{2\pi} \chi(k, i\omega) \mapsto T \sum_{n \in \mathbb{Z}} \chi(k, i\omega_n), \qquad \omega_n = 2\pi T n, \tag{27}$$

which yields

$$C(x,0) = T \sum_{n \in \mathbb{Z}} \int \frac{\mathrm{d}k}{2\pi} e^{-ikx} \chi(k, i\omega_n) = \frac{AT}{2c} \sum_{n \in \mathbb{Z}} \frac{e^{-\sqrt{\Delta^2 + \omega_n^2}\,|x|/c}}{\sqrt{\Delta^2 + \omega_n^2}} \overset{|x| \to \infty}{=} \frac{AT}{2c|\Delta|} e^{-|x||\Delta|/c}, \tag{28}$$

which we can match with Eq.(24) to fix [3]

$$A = 2cZ|\Delta|^{1/4}. \tag{29}$$

Notice that at the quantum critical point, A vanishes: this makes sense because elementary excitations on both sides of the QCP are of very different nature, so when entering the ordered phase there is zero overlap ($A = 0$) between the elementary excitations there and the spin-flip excitations of the paramagnetic phase. Also notice that the normalization $\hat{\sigma}^z \sim \sqrt{A}$ together with $\dim[\Delta] = +1$ recover the anomalous dimension $\dim[\hat{\sigma}^z] = 1/8$.

5.3 From static to dynamical at $T = 0$

Now let us compute the dynamical correlator at zero temperature, $K(x,t)$. This calculation is in fact completely analogous to that outlined in Eq.(26), only with an additional $e^{i\omega t}$ factor that turns into an $e^{i\epsilon_k t}$. One finds

$$K(x,t) = \frac{1}{2} \int \frac{\mathrm{d}k}{2\pi} e^{i\epsilon_k t} e^{-ikx} \frac{A}{\epsilon_k} = \frac{A}{2\pi c} \mathsf{K}_0(|\Delta| \sqrt{x^2/c^2 - t^2}). \tag{30}$$

Replacing Eq.(29) and looking at the classical (= non-relativistic) limit $x \ll ct$, this is

$$K(x,t) \overset{x \ll ct}{=} Z|\Delta|^{1/4} e^{it|\Delta|} \left(i2\pi t|\Delta|\right)^{-1/2} e^{i|\Delta|x^2/2c^2 t}, \tag{31}$$

the Feynman propagator of a particle with mass $|\Delta|/c^2$.

By contrast, in the other limit $x \gg ct$ of course one just recovers $C(x,0)$ obtained earlier in Eq.(28).

5.4 From $T = 0$ to finite temperature

Now finally we will build on $K(x,t)$, Eq.(30), to get the finite temperature correlator $C(x,t)$. While $K(x,t)$ describes the ballistic motion of a free quasiparticle, the physical difference at finite temperature is that this particle now collides with other (thermally excited) particles. The picture is in fact very similar to what we discussed in Sec.4.3 and depicted in Fig.5. The physics, however, is different. Each time our propagating particle (the black dashed line) collides with another thermally excited particle (a red lifeline), this multiplies the wavefunction with one insertion of the S-matrix which, as we have seen, is just a (-1) factor – just like two fermions crossing paths. That the two particles (the dashed black line and the solid red line) then continue straight on their trajectory, as we discussed in Sec.5.1, is a fortunate consequence of integrability in $d = 1$.

For a number n of crossings, we thus get the very same $(-1)^n$ factor as in Sec.4.3, but for a completely different reason. There, it was due to the red lines representing domain walls, that by definition flip the ground state orientation $\pm z$; here, it's due to the red lines representing spin-flips along $-x$ that are hard-core bosons, which in $d = 1$ just like fermions acquire a minus sign when they collide. In any case, mathematically the very same calculation carries over without a change, and the effect of collisions is the very same factor $R(x,t)$ that we had computed! Whence our final result in the disordered phase:

$$C(x,t) = K(x,t)\, R(x,t). \tag{32}$$

This formula indicates, physically, a clear separation of scales:

[3] Readers of this footnote should feel special: this derivation is not in Subir Sachdev's book and was made for your eyes only!

- Quantum fluctuations are in $K(x,t)$, complex oscillatory, with time scale Δ and spatial scale Δ/c;
- Thermal relaxation of correlations is in $R(x,t)$, a real smooth function, with time scale τ_φ and spatial scale ξ fixed by *classical* evolution.

Note, finally, that $\xi/c\tau_\varphi = (2T/\pi|\Delta|)^{1/2} \ll 1$: this is another separation of scales, telling us that the relaxation of correlations is much slower along the time axis than along the position axis. We can thus replace $R(x,t) \to R(0,t)$ in the finite-temperature structure factor,

$$S(k,\omega) = \int \mathrm{d}t \int \mathrm{d}x\, K(x,t)\, R(x,t) e^{i(\omega t - kx)} \approx \frac{A}{\epsilon_k} \frac{\tau_\varphi^{-1}}{(\omega-\epsilon_k)^2 + \tau_\varphi^{-2}}, \tag{33}$$

valid at small $k \ll \sqrt{T|\Delta|}/c$. Together with $S(k,\omega) \approx \frac{2T}{\omega}\,\mathrm{Im}\chi(k,\omega)$ (this is the $\omega \ll T$ version of fluctuation-dissipation), this latter form shows us that, to a very good approximation, the leading effect of collision-induced thermal relaxation is to broaden the free-particle result (applying fluctuation-dissipation to Eq.(25)) with a Lorentzian along the energy axis.

6 The quantum critical region

6.1 Basic correlator properties

What defines the quantum critical region is that $\lambda_{\mathrm{dB}} \sim \xi$, so there is no classical gas picture, and $\tau_\varphi^{-1} \sim T$, so there is no separation of scales between quantum and thermal fluctuations. Can we still say anything? Well deep in the critical region, $T \gg |\Delta|$, we may as well take $\Delta = 0$, that will be extremely useful all through this section. Then the model has Lorentz invariance, which implies that we can get $C(x,0)$ and $C(x,t)$ for the same price. Because we know the critical anomalous dimension $\dim[\hat\sigma^z] = 1/8$, the only possibility at $T = 0$ consistent with relativistic invariance is

$$K(x,-i\tau) \propto \left(\tau^2 + x^2/c^2\right)^{-1/8}, \qquad T = 0. \tag{34}$$

Equivalently, upon Fourier transform and analytic continuation to real frequencies this means the $T = 0$ dynamical susceptibility reads

$$\chi(k,\omega) \propto \left(c^2k^2 - (\omega + i0)^2\right)^{-7/8}, \qquad T = 0. \tag{35}$$

Instead of having poles, it has a full branch cut. This indicates that the critical operator $\hat\sigma^z$ has no overlap with any quasiparticle-like excitation (especially not those of the phases we investigated in Secs.4 and 5) , but instead behaves in a completely continuum-like way.

6.2 From zero to finite T using the conformal map

We have already noticed that the critical field theory Eq.(11) at $\Delta = 0$ has, in addition to Lorentz invariance, scale invariance with $z = 1$. In fact it enjoys full *conformal* invariance[4], and this in fact implies a very powerful way to obtain finite-temperature correlations directly from the zero-temperature counterparts. Indeed, for our theory Eq.(11) we have very easily computed *fermion* correlations both at zero and finite temperature, Eqs.(12a)–(12b). We see that one can be obtained from the other by means of the so-called conformal map:

$$c\tau \pm ix \quad \leftrightarrow \quad \frac{c}{\pi T}\sin\left((\pi T/c)(c\tau \pm ix)\right). \tag{36}$$

Why this map does not work only for free fermions described by Eq.(11), but instead for all correlations in any $D = 1+1$ conformal field theory, goes beyond the scope of this lecture. Here we will just use it as a tool.

Notice that applying the conformal map to $K(x,0)$ in Eq.(34) gives us

$$C(x,0) \propto T^{1/4} e^{-\pi T|x|/4c}, \tag{37}$$

[4]Although this is not directly a *consequence* of combined Lorentz and scale invariance (there is a note by Polchinski about that), it's typically included in the package.

from which we read $\xi^{-1} = \pi T/4c$. This recovers the exact asymptotic result Eq.(13), with $F(0) = \pi/4$. Here, not only did we get it basically for free, but scale invariance ensures this result holds for any x not just $|x| \to \infty$.

Of course, what we can't get just from invariance is the exact prefactor in Eq.(37), but this one is provided by Eq.(13). Reciprocally, this tells us that the exact prefactor in Eq.(34) is $Z\,G(0)\,(T/2\pi)^{1/4}$. Then we can just apply the conformal map to the dynamical $T = 0$ correlation $K(x, -i\tau)$ of Eq.(34), and we get

$$C(x, -i\tau) = Z\,(T/2)^{1/4}\,G(0)\left(\sin[\pi T(\tau - ix/c)]\,\sin[\pi T(\tau + ix/c)]\right)^{-1/8}, \tag{38}$$

that we are free to continue back to real times, $-i\tau \to t$. This is the universal result at $\Delta \ll T$, that holds not only asymptotically but for all x and t.

Let us once again marvel at the fact that conformal invariance, via the combined application of Lorentz transformations and the conformal map, has just enabled us to reconstruct the full exact dynamical correlator $C(x, t)$ just from the $|x| \to \infty$ asymptotics of the static correlator $C(x, 0)$.

Problem sheet – quantum phase transition of free fermions

Let us consider the following Lagrangean for free fermions in d dimensions, in euclidean notations

$$L = \psi^\dagger \partial_\tau \psi + (1/2m)\left|\partial_x \psi\right|^2 - \mu\,|\psi|^2. \tag{39}$$

[1] What is the quantum critical point of L? What phases does it separate? What is its dynamical exponent z? Why does L describe a universal critical theory?

[2] Write (as a momentum integral that you need not compute) the $T > 0$ correlator

$$G(x,t) = \langle \psi(x,t)\, \psi^\dagger(0,0)\rangle \tag{40}$$

and the fermionic free energy $\mathcal{F}$.

[3] Consider first $\mu < 0$ and $T \ll |\mu|$. Justify that the gas of fermions is essentially classical. Show that $G(x,t)$ is, at $T = 0$ and $\mu = 0$, the classical Feynman propagator Eq.(31), and that $\mathcal{F}$ is the free energy of a classical gas.

[4] Consider then $\mu > 0$ and $T \ll |\mu|$. At which scales (x,t) does the finite density of fermions start showing effects on particle propagation? From now on we specialize in $d = 1$. Having defined the Fermi momentum $k_{\rm F}$, linearize the theory near $\pm k_{\rm F}$, writing in particular $\psi(x,t) = e^{\pm i k_{\rm F} x}\psi_{\rm R/L}(x,t)$, and derive the effective Lagrangean

$$L_{\rm eff} = \psi_{\rm R}^\dagger\big(\partial_\tau - i v_{\rm F}\partial_x\big)\psi_{\rm R} + \psi_{\rm L}^\dagger\big(\partial_\tau + i v_{\rm F}\partial_x\big)\psi_{\rm L}. \tag{41}$$

What is the dynamical exponent z? Show that the reduced propagator is

$$G_{\rm R/L}(x,-i\tau) = \frac{T/2v_{\rm F}}{\sin\big(\pi T[\tau \pm i x/v_{\rm F}]\big)}. \tag{42}$$

Comment on the form of the result. What is the correlation length?

[5] Consider $T \gg |\mu|$. This is the critical region: hoes is that reflected in the typical length scales of the problem? Compute $G(x,0)$ at large $|x|$ and show $\xi = (m\pi T)^{-1/2}$. Argue that both quantum and thermal effects appear in this result. Looking at z, argue that this region is "controlled by" the quantum critical point. Compute $G(0,t)$ at large $|t|$. Comment on the behavior of correlations along the space and time axes.

[6] Establish the (μ, T) diagram of the system.

Solution sheet – quantum phase transition of free fermions

[1] The QCP sits at $\mu = 0$. It separates a dilute Fermi gas at $\mu < 0$ and a dense Fermi fluid at $\mu > 0$. To get the dynamical exponent, take $\mu = 0$ then count ∂_τ and ∂_x derivatives: one has $z = 2$. Still at $\mu = 0$, the theory is invariant upon rescaling $x' = xe^{-\ell}, \tau' = \tau e^{-z\ell}$ and $\psi' = \psi e^{d\ell/2}$. In fact $\dim[\mu] = 2$: it's the relevant parameter defining the QCP. The status of L as a universal critical theory comes from its stability against perturbations. Try adding an interaction term: Pauli exclusion imposes the presence of derivatives, the simplest possibility would be $\lambda(\psi^\dagger\partial_x\psi^\dagger)(\psi\partial_x\psi)$, and this has $\dim[\lambda] = -d$: perturbations are irrelevant.

[2] The propagator simply reads

$$G(x,t) = \int \frac{\mathrm{d}^d k}{(2\pi)^d} e^{ikx} \frac{e^{-i(k^2/2m-\mu)t}}{1+e^{-(k^2/2m-\mu)/T}} \tag{43}$$

and the free energy reads

$$\mathcal{F} = -T\int \frac{\mathrm{d}^d k}{(2\pi)^d} \ln\left(1+e^{-(k^2/2m-\mu)/T}\right). \tag{44}$$

[3] Compare $\lambda_{\mathrm{dB}} \sim T^{-1/2}$ with the inter-particle distance $l \sim e^{|\mu|/Td}$: clearly $l \gg \lambda_{\mathrm{dB}}$ so the gas is classical to a very good approximation. In $G(x,t)$ the denominator becomes just 1, and we get

$$G(x,t) \simeq (mT/2\pi)^{d/2} e^{-imx^2/2t} \tag{45}$$

which is the classical Feynman propagator (for a single particle). Similarly, approximating $\ln(1+\alpha) \approx \alpha$ for very small α, one obtains

$$\mathcal{F} \simeq -T(mT/2\pi)^{d/2} e^{-|\mu|/T} \tag{46}$$

the free energy of a classical Maxwell-Boltzmann gas.

[4] Even though L describes a free theory, the fermions still "see each other" via Pauli exclusion, whence the formation of a Fermi sea at $\mu > 0$, with size $k_{\mathrm{F}} = \sqrt{2m\mu}$. Notice that for $|x| \ll k_{\mathrm{F}}^{-1}$, the momentum integral in Eq.(43) gets most of its contributions from very large values $|k| \gg k_{\mathrm{F}}$ such that the denominator is 1. If one further assumes $|t| \ll 1/\mu$, the above $G(x,t)$ Eq.(45) still applies: basically a particle traveling a short distance and not taking much time to roam around does not get the chance to encounter any other fermion, so it does not realize there are other particles around and the free classical propagator still works.

Beyond these limits, it matters that we now have a Fermi sea. Write the Ansatz indicated in the problem sheet, and expanding, keep only those terms that are smallest order in $\partial_x\psi_{\mathrm{R/L}}$. This means, drop terms with $\partial^2_{xx}\psi_{\mathrm{R/L}}$ but keep those with $\partial_x\psi_{\mathrm{R/L}}\,\partial_x\left(e^{\pm ik_{\mathrm{F}}x}\right)$. This yields L_{eff}, Eq.(41), where $v_{\mathrm{F}} = k_{\mathrm{F}}/m$. One directly reads $z = 1$. Also note $\dim[v_{\mathrm{F}}] = 0$: again, velocity is a scale-invariant parameter of the theory. Now, writing $k = k_{\mathrm{F}} + p$, one has simply

$$G_{\mathrm{R/L}}(x,-i\tau) = \int \frac{\mathrm{d}p}{2\pi} \frac{e^{ip(\pm x + v_{\mathrm{F}} i\tau)}}{1+e^{-v_{\mathrm{F}}p/T}} = \frac{T/2v_{\mathrm{F}}}{\sin(\pi T(\tau \mp ix/v_{\mathrm{F}}))}. \tag{47}$$

The second equality comes from the contour integration, identifying poles $p_n = i(2n+1)\pi T/v_{\mathrm{F}}$, then applying the residue theorem and computing the (geometrical) sum. Notice that we have already encountered that form: one could have first calculated the $T = 0$ propagator then used the conformal map. This is because our effective theory Eq.(41) enjoys conformal invariance – this is not directly a consequence of its being a scale-invariant Lorentz-invariant theory with $z = 1$, although that's definitely a clue. Clearly, from this result, $\tau_\varphi = 1/T$ and $\xi = v_{\mathrm{F}}/T$.

[5] Just take Eq.(43) at $\mu = 0$, the particle density reads $\langle\psi^\dagger\psi\rangle = (2mT)^{d/2}\int\frac{\mathrm{d}^d\kappa}{(2\pi)^d}(1+e^{\kappa^2})^{-1}$, from which we read the inter-particle distance $l = (2mT)^{-1/2}$. Thus $l \sim \lambda_{\mathrm{dB}}$: one is indeed in the critical region where no scale separation is available and both quantum and thermal effects are important. Let us look at the static correlator

$$G(x,0) = \int\frac{\mathrm{d}k}{2\pi}\frac{e^{ikx}}{1+e^{-k^2/2mT}} \overset{|x|\to\infty}{\approx} -\pi(2mT)^{-1/2}e^{-(1-i)x\sqrt{\pi mT}}, \tag{48}$$

where we performed the integral again by identifying all the poles, applying the residue theorem, then at $|x| \to \infty$ only kept the pole closest to the real axis. This displays both an exponential decay and oscillations, that indicates both thermal and quantum effects appearing on the same scale. From this result, one reads $\xi^{-1} = \sqrt{\pi mT}$, again that same inverse length scale; its square root is consistent with $z = 2$ at the QCP (it yields $\dim[T] = 2$).

Finally, one can also look at the local correlator

$$G(0,t) = \int\frac{\mathrm{d}k}{2\pi}\frac{e^{-itk^2/2m}}{1+e^{-k^2/2mT}} \overset{|t|\to\infty}{\approx} e^{-i\pi/4}\left(m/8\pi t\right)^{1/2}, \tag{49}$$

computed this time using the stationary phase approximation. It displays algebraic correlations along the time axis. Obviously, this time the critical point is not conformal (as can be deduced from $z = 2$), indeed correlations behave differently as functions of x and t.